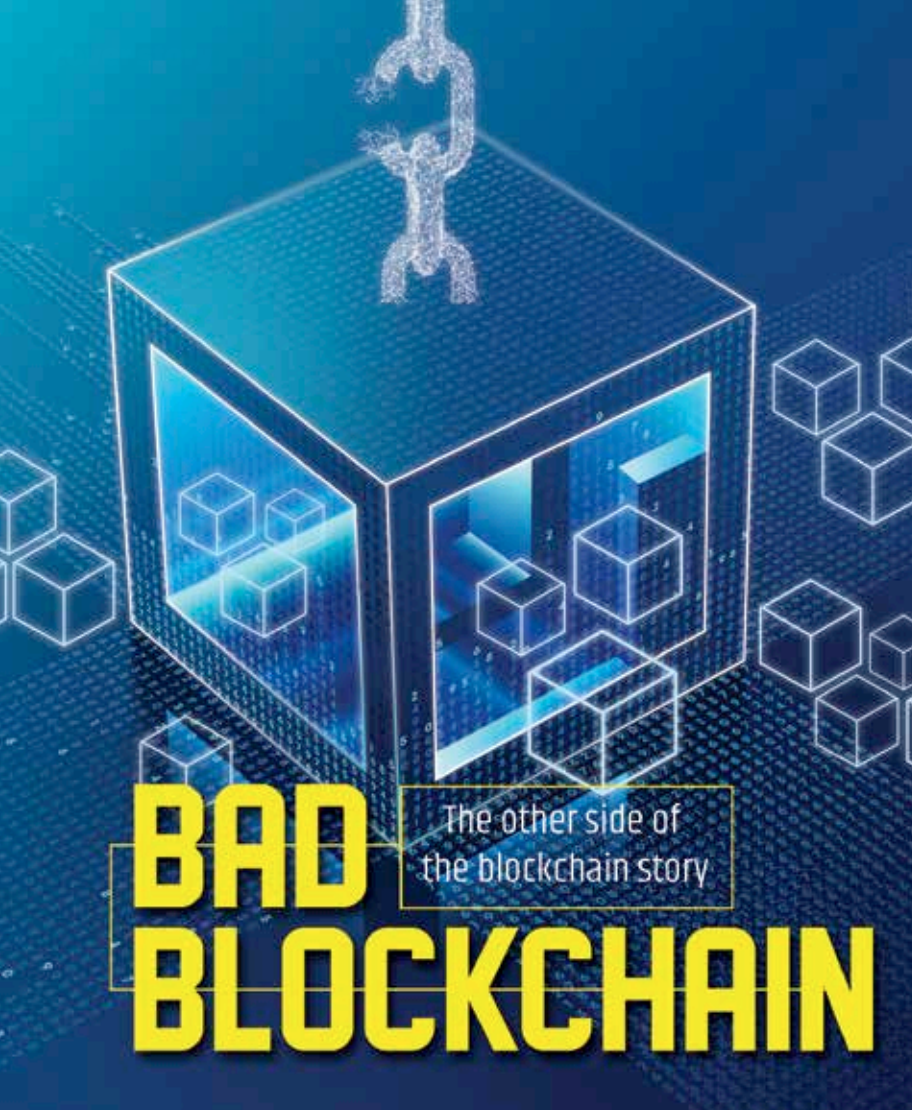




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Blockchain, like every hyped up piece of technology to surface over the past couple of years has garnered the support of numerous industries across various verticals. Blockchain has been touted to be a revolutionary technology, something that's destined to overcome the shortcomings of most ledger technologies. It has also seen some really strong criticism. Some of that is genuinely warranted while others lean more towards nitpicking. Let's take a look at why some proponents are vehemently opposed to blockchain in its current form.

NEW VULNERABLE

Blockchain is a new technology and that means that there hasn't been a lot of time spent towards understanding

where blockchain falters. Those with malicious intent or as we refer to them in common parlance – hackers – are still exploring the blockchain technology, trying to poke holes. Just in the last couple of months we've seen more news surfacing of breaches and attacks on blockchain enabled technology. Some of these include methods such as Vector76 attack, Finney attack, Race attack, Alternative history attack and the most popular of them all, Majority attack.

Everyone knows about Majority attack because the mechanism is ingrained in the way the blockchain



It's not as safe as you think.

algorithm is written. Referred to as the 51% attack, this mechanism relies on gaining control of 50% (or greater) of the network after which you have autonomy over rejecting or accepting transactions. This was one of the major concerns with Bitcoin since a group which controls more than 50% of the network effectively becomes the sole authority of the network. In 2014, the ghash.io pool briefly exceeded 50% of the mining power of the Bitcoin network.

Race attack is a sneaky one wherein an attacker sends a transaction to a merchant while sending a conflicting transaction to the rest of the pool. There's a greater likelihood of the conflicting transaction being accepted onto the ledger, thus, nullifying whatever was to be given to the merchant. Then there's the Finney attack method which works in a similar manner. To get more creative, hackers have combined the aforementioned attacks to achieve better results. While we used Bitcoins as an example here, these attacks are applicable to any application of a distributed ledger technology such as blockchain.

As the application of distributed ledger technologies (DLT) spreads to more domains, there will be more such vectors that are yet to be discovered. This is true for all new technologies, which is why we should devote more time to analysing DLT before building critical systems with it.

DISTRIBUTED NOT QUITE

Blockchain, in theory, is certainly a distributed system and we all know that if an entity should gain control of more than 50% of the network, then they gain the ability to double-spend. So keeping it decentralised is a way to prevent this exact thing from happening. However, in the real world, this is certainly not the case.

Take Bitcoins for example, the entire network is made up of pools. Each pool consists of multiple miners numbering in the hundreds or thousands or even millions. It's natural for people to come together as a pool because it makes it easy for people to mine and get paid on a regular basis. If not, then getting a block and then mining